La Scuola d'Italia / Huson / Physics Nickname:
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2.6 Classwork: Kinematics Review

Reference (provided). average speed $= \dfrac{\text{distance}}{\text{time}}$ $\bar{v} = \dfrac{\text{displacement}}{\text{time}}$ 1 h = 3600 s

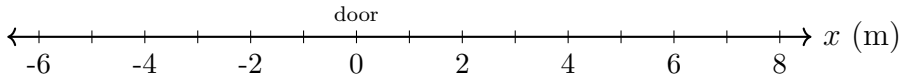
When the acceleration a is constant:

$$v = v_0 + at \qquad v^2 = v_0^2 + 2a(x - x_0)$$

$$x = x_0 + v_0 t + \tfrac{1}{2}at^2 \qquad \bar{v} = \frac{v + v_0}{2}$$

For a falling object, write y in place of x, and $a = g = 9.80$ m/s^2 downward. List what you know and what you want, then choose the equation. Show the units.

1. Take the classroom door as the origin, with east positive. A student walks 6.0 m east in 3.0 s, then 10.0 m west in 5.0 s. Draw each displacement as an arrow above the number line.



   (a) Where does the student end up? $x =$ _____

   (b) Distance walked = _____ Displacement = _____

   (c) Find the average speed and the average velocity for the whole walk.

2. In 2009 Usain Bolt set the 100 m world record, 9.58 s.

   (a) Find his average speed in m/s. Then convert it to km/h.

   (b) His top speed during the race was about 12.3 m/s. Why is his average speed lower?

3. A car is driving at 40 mi/h (17.9 m/s) when the driver brakes hard. In 2.3 you found that the car slows down at $a = -6.4$ m/s^2.

   (a) How long does the car take to stop?

   (b) Find the braking distance. The UK Highway Code gives 24 m at 40 mi/h. Does your answer agree?

4. A student drops a stone from a footbridge and sees it hit the water 2.00 s later. Take *down* as positive.

   (a) How fast is the stone going when it hits the water?

   (b) How high is the bridge above the water?

5. A ball is thrown straight up at 9.80 m/s. Take *up* as positive.

   (a) At the highest point, what is the ball's velocity? What is its acceleration?

   (b) How long does the ball take to reach the highest point?

   (c) How high does it go?

   (d) How long after the throw does it come back to the hand?